

[PUBLIC]

CI/CD Pipeline Standards — NovaTech Engineering

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This document defines the mandatory CI/CD pipeline stages, quality gates, and deployment strategies for all NovaTech services. GitHub Actions is the standard CI platform. ArgoCD manages GitOps-style continuous deployment to EKS. All pipelines must pass the security, contract, and performance gates before reaching production.

1. Pipeline Stages

Stage	Tool	Pass Criteria	SLA
Lint & Format	Ruff, Black	Zero violations	< 2 min
Unit Tests	pytest	Coverage >= 85%	< 5 min
Security Scan	Trivy, Bandit	No HIGH/CRIT CVEs	< 3 min
Contract Tests	Pact	All consumer pacts green	< 4 min
Build & Push	Docker, ECR	Image signed via Cosign	< 6 min
Perf Gate	k6	P95 < 500ms at 100 RPS	< 8 min
Deploy (Staging)	ArgoCD	Health checks green	< 10 min
Deploy (Prod)	ArgoCD	Canary 5%→25%→100%	< 20 min

2. Deployment Strategy

Production deployments use a canary strategy: 5% traffic for 10 minutes, then 25% for 10 minutes, then 100% if error rate remains below 0.5% and P95 latency stays below 500 ms. Automated rollback triggers if either threshold is breached during the canary phase.

Feature flags (LaunchDarkly) are used to decouple deploy from release. Project Atlas features default to off in production and are enabled per tenant by the Product team after internal testing.

3. Secrets Management in CI

No secrets may be stored in GitHub repository secrets for production environments. CI workflows use OIDC to assume an IAM role and fetch secrets from AWS Secrets Manager at runtime. Staging uses a dedicated IAM role with scoped permissions.